EX 1: MANUAL CALCULATION OF Y-BUS MATRIX

GIVEN DATA

Line 1 : Bus 1 – Bus 2

$Z12 = 0.10 + j0.40$ p.u.

Line Charging Admittance = $j0.015$ p.u.

Line 2 : Bus 2 – Bus 3

$Z23 = 0.15 + j0.60$ p.u.

Line Charging Admittance = $j0.020$ p.u.

Line 3 : Bus 3 – Bus 4

$Z34 = 0.18 + j0.55$ p.u.

Line Charging Admittance = $j0.018$ p.u.

Line 4 : Bus 4 – Bus 1

$Z41 = 0.10 + j0.35$ p.u.

Line Charging Admittance = $j0.012$ p.u.

Line 5 : Bus 4 – Bus 2

$Z42 = 0.25 + j0.20$ p.u.

Line Charging Admittance = $j0.030$ p.u.

SOLUTION:

STEP 1: CALCULATION OF LINE ADMITTANCES

$Y12 = 1/(0.10 + j0.40) = (0.10 - j0.40)/(0.10^2 + 0.40^2) = -0.5882 - j2.3529$ p.u.

$Y23 = 1/(0.15 + j0.60) = (0.15 - j0.60)/(0.15^2 + 0.60^2) = -2.3922+j1.5666$ p.u.

$Y34 = 1/(0.18 + j0.55) = (0.18 - j0.55)/(0.18^2 + 0.55^2) = -0.5378 - j1.6420$ p.u.

$Y41 = 1/(0.10 + j0.35) = (0.10 - j0.35)/(0.10^2 + 0.35^2) = -0.7547 + j2.6415$ p.u.

$Y42 = 1/(0.25 + j0.20) = (0.25 - j0.20)/(0.25^2 + 0.20^2) = -2.4390 + j0.9512$ p.u.

STEP 2 : OFF-DIAGONAL ELEMENTS

Y12 = Y21 = -0.5882+j2.3529

Y23 = Y32 = -2.3922+j1.5666

Y34 = Y43 = -0.5378 + j1.6420

Y13 = Y31 = -0.5882+j2.3529

Y24 = Y42 = 0

STEP 3 : DIAGONAL ELEMENTS

Y11 = Y12 + Y14 + (j0.015/2) + (j0.012/2)

= (0.5882 - j2.3529) + (0.7547 - j2.6415) + j0.0075 + j0.006 = 1.3429 - j4.9809

Y22 = Y12 + Y23 + Y24 + (j0.015/2) + (j0.020/2) + (j0.030/2)

= (0.5882 - j2.3529) + (0.3922 - j1.5686) + (2.4390 - j1.9512) + j0.0325 = 3.4194-j6.8602

Y33 = Y23 + Y34 + (j0.020/2) + (j0.018/2)

= (0.3922 - j1.5686) + (0.5378 - j1.6420) + j0.019 = 0.9980-j3.1960

Y44 = Y14 + Y24 + Y34 + (j0.012/2) + (j0.018/2) + (j0.030/2)

= (0.7547 - j2.6415) + (2.4390 - j1.9512) + (0.5378 - j1.6420) + j0.030 = 3.7313-j6.7070

FINAL Y-BUS MATRIX

[1.3429-j4.9809	-0.5882+j2.3529	-0.5882+j2.3529	0]
[-0.5882+j2.3529	3.4194-j6.8602	-2.3922+j1.5666	-0.7547+J2.9412]
[0	-0.3922+j1.5866	0.9980-j3.1960	-0.5378+j1.6420]
[-0.7547+j2.6415	-2.4390+j1.9512	-0.5378+j1.6420	3.7313-j6.7070]